Quantum Computing

Quantum Networks and Cryptography

Lecture 8 Notes

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About these notes

These notes accompany a nine-lecture course on quantum computing designed and produced by the Lebanese Science Organization. Each chapter corresponds to one lecture. The lectures are concept-focused: they explain what a quantum state is, how entanglement, interference, and measurement combine to produce computation, and what is required to make that computation reliable. Linear algebra, complex numbers, and elementary probability are assumed.

Each chapter names its lecturer and date, opens with a short preface, develops its material with the derivations in place, and closes with a summary. Worked examples appear in numbered boxes, as do definitions, theorems, remarks, and common misconceptions. Works cited in the text are collected in a single reference list at the end of the document.

Conventions used throughout: qubits in a register are labelled from left to right, so that in $|ab\rangle$ the first qubit is a; the first qubit is the top wire in every circuit diagram; the computational basis of two qubits is ordered $|00\rangle, |01\rangle, |10\rangle, |11\rangle$; circuits are read from left to right, so a gate U_1 followed by a gate U_2 acts as the matrix product U_2U_1.

This chapter is exported individually from the full lecture notes (8 of 9). These notes are made available for open, public study. Reuse conditions will be stated separately by the Lebanese Science Organization.

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Chapter 1

Quantum Networks and Cryptography

Lecturer: Marc Fares

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Entanglement can be distributed between distant nodes and extended across a network by local operations and classical messages, even though no device can copy or amplify a qubit in transit. This lecture explains what a quantum network provides, with entanglement swapping as its basic primitive, and presents quantum key distribution as the most developed application. It then turns to the other side of the same ideas: Shor's algorithm breaks the public-key schemes in wide use today, Grover's algorithm only trims the security of symmetric ciphers and hash functions, and post-quantum cryptography is the practical response.

1.1 Entanglement distribution

A classical network moves bits. A node receives a packet, reads it, copies it into a buffer, and forwards a copy. Every one of those steps copies information, and none of them is available for quantum information: the no-cloning theorem of **??** forbids copying an unknown qubit, and **??** showed that measuring a qubit disturbs it and returns a single classical outcome rather than its amplitudes. A network for quantum information must therefore be built on a different primitive.

That primitive is shared entanglement. A quantum network is a collection of nodes, each holding a small number of qubits in a quantum memory, connected by channels that can carry qubits, typically photons in optical fibre or free space, and by an ordinary classical network. The purpose of the quantum channels is to establish entangled pairs, in the ideal case Bell states $|\Phi^+\rangle$ of **??**, between the qubits of neighbouring nodes. The entangled pair, not the photon that produced it, is the resource. Once two nodes share $|\Phi^+\rangle$, they can teleport a qubit from one to the other using two classical bits, as in **??**, or they can consume the pair directly in a protocol such as the key distribution of section 1.2.

Loss and the failure of classical repeating. A photon sent through optical fibre survives with a probability that decays exponentially with distance; standard telecommunication fibre attenuates by about 0.2 decibels per kilometre, so roughly half the photons are lost every 15 kilometres. A classical fibre link solves the same problem with repeaters that detect the weakened signal and retransmit a fresh copy. This is exactly the measure-and-resend or amplify strategy that no-cloning rules out. A device that detected the photon's state and re-emitted it would have to measure, which yields one outcome in one basis and destroys the rest; a device that amplified the signal without

measuring would produce copies, which no linear map can do for unknown states. There is no quantum analogue of the classical measure-and-resend repeater, and as of 2025 direct transmission of entanglement through fibre without intermediate nodes is limited to a few hundred kilometres [1].

The way around this limitation is not to send entanglement further but to build long-distance entanglement out of short-distance entanglement [2].

Definition 1.1: Entanglement swapping

Let a left node hold qubit A, a middle node hold qubits B and C, and a right node hold qubit D, with A and B in the state $|\Phi^+\rangle$ and C and D in the state $|\Phi^+\rangle$. Entanglement swapping is the following procedure: the middle node measures B and C in the Bell basis, obtaining one of four outcomes, and sends the two-bit outcome to the right node over the classical network; the right node applies a Pauli correction to D determined by the outcome. Afterwards A and D are in the state $|\Phi^+\rangle$, although they have never interacted and have never been in the same place.

Proposition 1.1 The swapping identity

With qubits ordered A, B, C, D,

$$\begin{aligned}|\Phi^+\rangle_{AB} \otimes |\Phi^+\rangle_{CD} = \frac{1}{2}\Big(&|\Phi^+\rangle_{AD} \otimes |\Phi^+\rangle_{BC} + |\Phi^-\rangle_{AD} \otimes |\Phi^-\rangle_{BC} \\ &+ |\Psi^+\rangle_{AD} \otimes |\Psi^+\rangle_{BC} + |\Psi^-\rangle_{AD} \otimes |\Psi^-\rangle_{BC} \Big).\end{aligned} \tag{1.1}$$

A Bell measurement on B and C therefore yields each of the four outcomes with probability $\frac{1}{4}$ and leaves A and D in the Bell state carrying the same label. The corrections $\mathbb{I}$, Z, X, ZX on D, for the outcomes $|\Phi^+\rangle$, $|\Phi^-\rangle$, $|\Psi^+\rangle$, $|\Psi^-\rangle$ respectively, return A and D to $|\Phi^+\rangle$.

Equation (1.1) is the whole content of this section. The four-qubit state, entangled only across AB and across CD, is rewritten as a superposition of states entangled only across AD and across BC. Measuring BC in the Bell basis selects one term, and the classical message tells the right node which one. The reader who has seen teleportation in **??** will recognize the procedure: the middle node teleports its half of the AB pair, namely B, to the right node using the CD pair as its resource, and the entanglement with A travels with it. The full expansion, the circuit, and the correction table are given in section 1.1.1.

1.1.1 The swapping identity

Order the qubits A, B, C, D and expand the product of the two Bell pairs:

$$\begin{aligned}|\Phi^+\rangle_{AB} \otimes |\Phi^+\rangle_{CD} &= \frac{1}{2}\left(|00\rangle_{AB} + |11\rangle_{AB}\right) \otimes \left(|00\rangle_{CD} + |11\rangle_{CD}\right) \\ &= \frac{1}{2}\left(|0000\rangle + |0011\rangle + |1100\rangle + |1111\rangle\right).\end{aligned} \tag{1.2}$$

Regroup each term as a state of AD times a state of BC. In $|0000\rangle$ the pair AD is $|00\rangle$ and the pair BC is $|00\rangle$; in $|0011\rangle$ the pair AD is $|01\rangle$ and BC is $|01\rangle$; in $|1100\rangle$ the pair AD is $|10\rangle$ and BC is $|10\rangle$; in $|1111\rangle$ both pairs are $|11\rangle$. Hence

$$|\Phi^+\rangle_{AB} \otimes |\Phi^+\rangle_{CD} = \frac{1}{2} \sum_{j,k\in\{0,1\}} |jk\rangle_{AD} \otimes |jk\rangle_{BC}\,. \tag{1.3}$$

Now write the computational basis states of BC in the Bell basis, inverting **??**:

$$|00\rangle = \frac{|\Phi^+\rangle + |\Phi^-\rangle}{\sqrt{2}}, \qquad |11\rangle = \frac{|\Phi^+\rangle - |\Phi^-\rangle}{\sqrt{2}}, \qquad |01\rangle = \frac{|\Psi^+\rangle + |\Psi^-\rangle}{\sqrt{2}}, \qquad |10\rangle = \frac{|\Psi^+\rangle - |\Psi^-\rangle}{\sqrt{2}}. \tag{1.4}$$

Substituting into (1.3) and collecting the coefficient of each Bell state of BC,

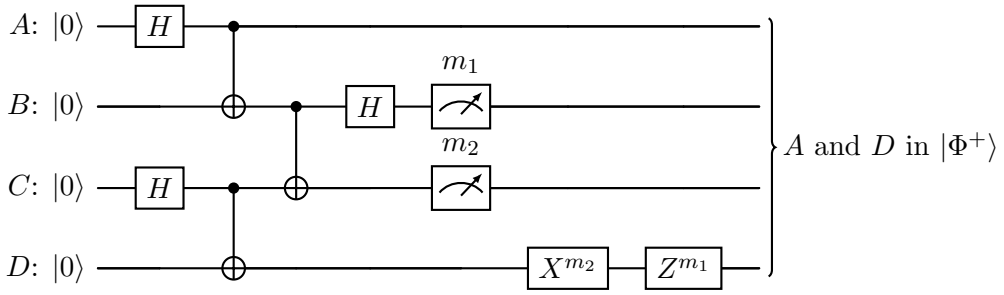
$$\begin{aligned} \left|\Phi^+\right\rangle_{AB} \otimes \left|\Phi^+\right\rangle_{CD} &= \frac{1}{2\sqrt{2}}\Big[(|00\rangle + |11\rangle)_{AD} \otimes \left|\Phi^+\right\rangle_{BC} + (|00\rangle - |11\rangle)_{AD} \otimes \left|\Phi^-\right\rangle_{BC} \\ &\qquad + (|01\rangle + |10\rangle)_{AD} \otimes \left|\Psi^+\right\rangle_{BC} + (|01\rangle - |10\rangle)_{AD} \otimes \left|\Psi^-\right\rangle_{BC} \Big] \\ &= \frac{1}{2}\Big(\left|\Phi^+\right\rangle_{AD} \otimes \left|\Phi^+\right\rangle_{BC} + \left|\Phi^-\right\rangle_{AD} \otimes \left|\Phi^-\right\rangle_{BC} \\ &\qquad + \left|\Psi^+\right\rangle_{AD} \otimes \left|\Psi^+\right\rangle_{BC} + \left|\Psi^-\right\rangle_{AD} \otimes \left|\Psi^-\right\rangle_{BC} \Big), \end{aligned} \tag{1.5}$$

which is (1.1). The four terms are orthogonal and have equal weight, so each Bell outcome on BC has probability $\frac{1}{4}$, and conditioned on the outcome the state of AD is the Bell state with the same label. The corrections on D are read off from the actions $Z|1\rangle = -|1\rangle$ and $X|0\rangle = |1\rangle$:

Outcome on BC	Measured bits (m_1, m_2)	State of AD	Correction on D
$\|\Phi^+\rangle$	$(0,0)$	$\|\Phi^+\rangle$	$\mathbb{I}$
$\|\Psi^+\rangle$	$(0,1)$	$\|\Psi^+\rangle$	X
$\|\Phi^-\rangle$	$(1,0)$	$\|\Phi^-\rangle$	Z
$\|\Psi^-\rangle$	$(1,1)$	$\|\Psi^-\rangle$	ZX, that is X then Z

For instance $(\mathbb{I}\otimes X)|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle) = |\Phi^-\rangle$ and then $(\mathbb{I}\otimes Z)|\Phi^-\rangle = |\Phi^+\rangle$. The measured bits follow the convention of **??**, in which the Bell measurement is the preparation circuit run backwards: a CNOT with B as control and C as target, then H on B, then a computational-basis measurement of both, so that $|\Phi^+\rangle \mapsto 00$, $|\Psi^+\rangle \mapsto 01$, $|\Phi^-\rangle \mapsto 10$, $|\Psi^-\rangle \mapsto 11$. In that convention the correction is $Z^{m_1}X^{m_2}$ on D, exactly as in teleportation.

The complete circuit, including the preparation of the two Bell pairs, is the following. The first two gates on each pair prepare $|\Phi^+\rangle$; the CNOT, the H, and the two meters on B and C form the Bell measurement just described.



In a network the two preparations happen at different places and times, the measured bits travel over the classical network, and the qubits A and D wait in quantum memories until the correction is applied. The correction may equally be applied to A instead of D; only the classical destination changes.

1.1.2 Quantum repeaters

Swapping alone does not defeat loss, since each short link still fails most of the time. Two further ingredients turn it into a scalable scheme [3, 1].

Definition 1.2: Quantum repeater

A quantum repeater is a network node that combines three capabilities: a quantum memory, which stores a qubit for long enough that entanglement on the neighbouring links can be attempted repeatedly until both succeed; entanglement swapping, which joins the two successful links into one longer one; and entanglement purification, which consumes several imperfect entangled pairs and produces fewer pairs of higher fidelity (a measure of how close a prepared state is to the intended one), using only local operations and classical communication. A chain of repeaters extends entanglement over a distance at which direct transmission would almost never succeed.

The three components are listed as requirements, not as constructions, and each is an active research problem. The point of the definition is the logic: loss is defeated by heralding, in which a node waits for a classical signal that a link has succeeded rather than trying to amplify the photon; distance is defeated by swapping; imperfection is defeated by purification. The quantitative theory of how many repeaters and how good a memory a given distance and rate require is outside the course.

Remark 1.1 Classical and quantum networks

Nothing in this section moves classical data faster or in greater volume. Every protocol on a quantum network still uses the classical network to carry measurement outcomes, heralding signals, and corrections, and **??** guarantees that the entanglement itself carries no message. What a quantum network adds is a resource that the classical network cannot provide at all: shared entanglement between distant parties. Its applications, which the next section surveys, are tasks that are either impossible or insecure with classical communication alone, not classical tasks performed faster.

1.2 Quantum key distribution

The most developed application of quantum communication is the distribution of secret keys. The problem it solves is old. Two parties who share a secret random string can communicate with unconditional secrecy by the one-time pad; the difficulty is obtaining that shared string when every channel between them may be observed. Quantum mechanics offers a solution built on two facts: nonorthogonal states cannot be distinguished with certainty, as the failure of a computational-basis measurement to tell $|+\rangle$ from $|0\rangle$ illustrates, and a measurement in the wrong basis disturbs the state. An eavesdropper who wants to learn the key must interact with the qubits, and that interaction leaves a statistical trace.

Definition 1.3: Quantum key distribution

A quantum key distribution protocol allows two parties, Alice and Bob, connected by an insecure quantum channel and an authenticated classical channel, to agree on a shared random key such that any eavesdropper who has interacted with the quantum channel either is detected, in which case the key is discarded, or is left with a negligible amount of information about the key.

1.2.1 The BB84 protocol

Alice prepares a sequence of qubits [4]. For each position she chooses a random bit and a random basis, the computational basis $\{|0\rangle, |1\rangle\}$ or the Hadamard basis $\{|+\rangle, |-\rangle\}$, and sends the corresponding state, encoding 0 as $|0\rangle$ or $|+\rangle$ and 1 as $|1\rangle$ or $|-\rangle$. Bob measures each qubit in a basis of his own random choice and records the outcome. When the transmission is complete, Alice and Bob announce their basis choices, but not their bits, over the classical channel and discard every position at which the bases differ; this step is called sifting and keeps about half the positions. At the kept positions, in the absence of noise and eavesdropping, Bob's outcome equals Alice's bit with certainty. Alice and Bob then sacrifice a random sample of the sifted positions, compare the bits openly, and estimate the error rate. If it is low enough they proceed to classical postprocessing, described in section 1.2.4, and obtain a shared key; if it is too high they abort.

The protocol is secure because Eve does not know the basis of any qubit while it is in transit. The simplest attack, in which Eve measures each qubit in a basis of her own choice and sends Bob a fresh qubit in the state she found, introduces errors at one quarter of the sifted positions in the ideal case, which the sample comparison detects. The one-line reason is that in half the positions Eve guesses the wrong basis, and in those positions Bob's bit is a fair coin. The derivation is in section 1.2.3. Full security proofs cover all attacks permitted by quantum mechanics, including coherent attacks on many qubits at once, and are outside the course. Two classical requirements remain in every version of the protocol. The classical channel may be public, but it must be authenticated, since otherwise Eve can run one instance of the protocol with each party and relay between them, as section 1.2.5 explains; authentication needs a short pre-shared key or a classical signature, so quantum key distribution expands a small secret into a large one rather than creating one from nothing. And the sifted string is not yet a key: it must pass through parameter estimation, information reconciliation, and privacy amplification, described in section 1.2.4.

1.2.2 Example run

The table shows ten positions. Alice's basis is Z for the computational basis and X for the Hadamard basis; the state column is what travels. Bob's outcome is what he records; where his basis differs from Alice's, his outcome is a fair coin and the position is discarded at sifting. No eavesdropper is present.

Position	1	2	3	4	5	6	7	8	9	10
Alice's bit	0	1	1	0	1	0	0	1	1	0
Alice's basis	Z	X	Z	X	X	Z	X	Z	X	Z
State sent	$\|0\rangle$	$\|-\rangle$	$\|1\rangle$	$\|+\rangle$	$\|-\rangle$	$\|0\rangle$	$\|+\rangle$	$\|1\rangle$	$\|-\rangle$	$\|0\rangle$
Bob's basis	Z	Z	Z	X	Z	X	X	Z	X	X
Bob's outcome	0	1	1	0	0	1	0	1	1	0
Bases agree	yes	no	yes	yes	no	no	yes	yes	yes	no
Sifted key	0		1	0			0	1	1	

Six of the ten positions survive sifting, close to the expected half, and at every surviving position Bob's outcome equals Alice's bit. At positions 2, 5, 6, and 10 Bob measured in the wrong basis; his outcomes there happen to agree with Alice's bit twice and to disagree twice, which is the fair-coin behaviour that makes these positions useless. Alice and Bob would next reveal, say, positions 1 and 8 to estimate the error rate, finding none, and keep positions 3, 4, 7, and 9 as raw key.

1.2.3 The intercept-resend attack

Eve intercepts each qubit, measures it in a basis chosen at random from $\{Z, X\}$, and sends Bob a fresh qubit in the state corresponding to her outcome in her basis. Consider a sifted position, at which Alice's and Bob's bases agree. With probability $\frac{1}{2}$ Eve's basis also agrees. Then her measurement returns Alice's bit with certainty, since the state is a basis vector of the basis she measures, and she resends exactly the state Alice sent; Bob measures it correctly and no error is introduced. With probability $\frac{1}{2}$ Eve's basis differs from Alice's. Then Alice's state is $|+\rangle$ or $|-\rangle$ when Eve measures in Z, or $|0\rangle$ or $|1\rangle$ when Eve measures in X, and in either case Eve obtains each outcome with probability $\frac{1}{2}$ and learns nothing. She resends a basis vector of the wrong basis, for example $|0\rangle$ when Alice sent $|+\rangle$. Bob, measuring in Alice's basis, then obtains each outcome with probability $\frac{1}{2}$, so his bit is wrong with probability $\frac{1}{2}$. The sifted error rate is therefore

$$\Pr[\text{error}] = \tfrac{1}{2} \cdot 0 + \tfrac{1}{2} \cdot \tfrac{1}{2} = \tfrac{1}{4}. \tag{1.6}$$

Comparing a sample of n sifted bits, Alice and Bob would see about $n/4$ disagreements where an undisturbed channel gives none. Eve learns the correct bit only at the half of the positions where her basis agreed, so she gains partial information at the price of a large disturbance. Gentler attacks produce less disturbance and less information; the security proofs quantify this trade-off for every attack that quantum mechanics permits.

1.2.4 Classical postprocessing

Three classical steps follow the quantum exchange. Parameter estimation is the comparison of a random sample of sifted bits over the authenticated channel to estimate the error rate; the sample is then discarded, and the protocol aborts if the error rate exceeds a threshold fixed in advance by the security analysis. Information reconciliation is an error-correction step in which Alice and Bob exchange parity information over the public channel until their remaining strings agree; every bit exchanged is assumed known to Eve and is accounted for. Privacy amplification is the application of a publicly chosen hash function that compresses the reconciled string to a shorter one, chosen so that Eve's information about the result, which was bounded by the estimated error rate and by the reconciliation leakage, becomes negligible. The output is the key. Each step is classical, and the final key length is smaller than the sifted length by an amount determined by the error rate.

1.2.5 Authentication

Suppose the classical channel carries no authentication. Eve cuts both channels and impersonates Bob to Alice and Alice to Bob, running one complete instance of the protocol with each. She ends with a key shared with Alice and a different key shared with Bob, and can decrypt everything Alice sends, read it, and re-encrypt it for Bob. Neither party detects anything, since each instance of the protocol was run correctly. The quantum channel prevented undetected eavesdropping; it did not establish who was at the other end. Authentication of the classical messages, using a short pre-shared secret or a classical signature scheme, is therefore a precondition of every quantum key distribution protocol.

1.2.6 Entanglement-based key distribution

A quantum network delivers Bell pairs, and a Bell pair is a key-distribution device. If Alice and Bob share $|\Phi^+\rangle$ and each measures in a random choice of the computational or Hadamard basis, then **??** tells us that whenever the bases agree the outcomes agree, and the sifted outcomes are random bits that neither party chose. This is the protocol of Bennett, Brassard, and Mermin (1992) [5], and it is equivalent to BB84 with Alice's preparation replaced by her measurement of one half of

the pair. Ekert's earlier protocol (1991) [6] uses additional measurement settings and a CHSH test, as in **??**, to certify the entanglement directly. The entanglement-based view makes the role of the network explicit: whoever can deliver high-fidelity Bell pairs between two nodes can deliver keys between them, and the repeaters of section 1.1 exist precisely to deliver such pairs over distances that a photon cannot cross.

1.2.7 Other applications

Three further uses of a quantum network are worth naming, each in one sentence. Distributed quantum computing uses teleportation between nodes to run a circuit whose qubits are spread across several processors, at the cost of one Bell pair and two classical bits per teleported qubit. Networked sensing shares entanglement between separated sensors, such as clocks or telescopes, so that the phase estimation ideas of **??** apply to a quantity defined over the whole array rather than at one point. Blind quantum computation lets a client with a small quantum device delegate a computation to a server so that the server learns neither the input, the output, nor the program. None of these is developed further in this course.

1.3 Cryptographic consequences

Modern cryptography rests on two kinds of primitive. Public-key schemes, in which the key used to encrypt or verify is public and the key used to decrypt or sign is private, are what allows two strangers to establish a shared secret over the internet and what authenticates software updates and web servers. Symmetric schemes, in which both parties hold the same key, are what encrypts the bulk of the data once a shared secret exists; hash functions, which map arbitrary data to a fixed-length digest, are used for integrity, commitments, and signatures. Quantum algorithms affect the two kinds very differently.

Public-key cryptography. The public-key schemes in wide use derive their security from two problems. RSA is secure only if factoring the product of two large primes is infeasible. The Diffie-Hellman key exchange, the Digital Signature Algorithm, and their elliptic-curve versions are secure only if the discrete logarithm problem (the analogue of factoring on which Diffie-Hellman and elliptic-curve schemes rest) is infeasible. Both problems are instances of the hidden-period structure that **??** exploited: Shor's algorithm solves factoring and discrete logarithms in a number of operations polynomial in the key length, whereas the cost of the best classical algorithms grows faster than any polynomial in the key length. A quantum computer large and reliable enough to run Shor's algorithm on a 2048-bit RSA modulus or a 256-bit elliptic curve breaks these schemes outright. This is not a reduction of security margin that can be absorbed by choosing longer keys, since the algorithm's cost grows only polynomially with the key length.

Symmetric ciphers and hash functions. Symmetric ciphers such as AES, and hash functions such as SHA-2 and SHA-3, have no periodic structure for Shor's algorithm to exploit. The only generic quantum attack is the search of **??**: Grover's algorithm finds a key among 2^n candidates in about $2^{n/2}$ evaluations of the cipher, and the BBBV bound shows that no quantum algorithm does better without exploiting structure. Two facts limit the consequences. First, the speedup is quadratic, so it halves the effective key length rather than destroying the scheme: AES-128 retains about 64 bits of security against a sequential Grover search, and more in practice because the search parallelizes poorly, and AES-256 about 128 bits. Second, the speedup is poorly parallelizable. Splitting a Grover search among k machines reduces the time by a factor of about $\sqrt{k}$, not k, so the 2^{64} sequential quantum evaluations for AES-128 cannot be traded for many short parallel runs in the way a classical brute-force search can. On this basis the NIST Post-Quantum Cryptography project

states that AES-128 is expected to remain secure for the foreseeable future and that AES-256 is comfortably beyond reach [7]; SHA-256 and SHA-3 are in the same position. Doubling key lengths where convenient is prudent, but symmetric cryptography is not what needs to be replaced.

Wrong Concept 1.1: Quantum computers will break all encryption

A large quantum computer would break RSA, Diffie-Hellman, and elliptic-curve cryptography, because Shor's algorithm solves the exact problems on which they rest. It would not break AES or the standard hash functions, because the only generic quantum attack on them is Grover's search, whose quadratic and poorly parallelizable speedup leaves AES-128 secure and AES-256 far out of reach. Since almost all encrypted traffic is carried by symmetric ciphers, with public-key methods used only to exchange the symmetric key and to sign, the correct statement is that quantum computers threaten the key exchange and the signatures, and that those specific components can be, and are being, replaced.

1.3.1 Affected and unaffected schemes

The table summarizes the effect of the two algorithms of **??** and **??** on the principal primitives. Effective security is expressed as the base-two logarithm of the number of quantum operations of the best known generic attack.

Primitive	Hard problem	Algorithm	Effect at sufficient scale
RSA	Integer factoring	Shor	Broken in polynomial time
Diffie-Hellman, DSA	Discrete logarithm in finite fields	Shor	Broken in polynomial time
Elliptic-curve DH and DSA	Elliptic-curve discrete logarithm	Shor	Broken in polynomial time
AES-128	Key search over 2^{128} keys	Grover	Effective security about 64 bits, sequential; assessed by NIST as secure for the foreseeable future
AES-256	Key search over 2^{256} keys	Grover	Effective security about 128 bits; comfortably beyond reach
SHA-256	Preimage search over 2^{256} digests	Grover	Preimage security about 128 bits; collision search gains less in practice; remains secure

The first three rows are what post-quantum cryptography replaces. The last three rows are what it keeps, with larger parameters where convenient.

1.3.2 Post-quantum cryptography

Harvest now, decrypt later. The threat to public-key cryptography is not only a future one. An adversary who records encrypted traffic today, together with the public-key exchanges that protected it, can decrypt it whenever a sufficiently large quantum computer becomes available. For secrets that must stay confidential for decades, such as state, medical, or industrial records, the relevant question is therefore not when such a machine will exist but whether it might exist within the lifetime of the secret. This is the argument that drives migration before any cryptographically relevant quantum computer has been built.

Definition 1.4: Post-quantum cryptography

Post-quantum cryptography consists of public-key schemes designed to run on ordinary classical hardware and to resist attacks by both classical and quantum computers. Their security rests on computational problems, principally in lattices and in hash functions, for which no efficient quantum algorithm is known. NIST published its first post-quantum standards in August 2024; they are classical lattice-based and hash-based schemes, listed in section 1.3.3.

These standards are classical mathematics. Deploying them requires new software, not new physics, and browsers and messaging protocols began shipping post-quantum key exchange in 2023, initially with draft Kyber parameters and, after August 2024, with ML-KEM; the OpenSSH remote-login software had already defaulted to a different lattice-based hybrid in 2022. Their security is a computational assumption, of the same kind as the assumption behind RSA, but with a different underlying problem; it is not a theorem of physics, and it could in principle be undermined by a new algorithm, classical or quantum. Post-quantum cryptography and the quantum key distribution of section 1.2 are often confused: the former runs on existing hardware, provides both key exchange and signatures, and rests on a computational assumption, whereas the latter requires a quantum channel, provides key agreement only, and rests on physics together with classical authentication. Standardization bodies have accordingly chosen post-quantum cryptography as the primary response and treat quantum key distribution as a complementary option for specific links; the point-by-point comparison is Remark 1.2 in section 1.3.3.

1.3.3 Standardized algorithms

In August 2024 NIST published three post-quantum standards: FIPS 203, the lattice-based key-encapsulation mechanism ML-KEM [8]; FIPS 204, the lattice-based signature scheme ML-DSA [9]; and FIPS 205, the hash-based signature scheme SLH-DSA [10]. ML-KEM is the key exchange that browsers and messaging protocols adopted after August 2024, replacing the draft Kyber parameters that the earliest deployments of 2023 and early 2024 had used. All three run on ordinary classical hardware and rest on computational assumptions, in lattices for the first two and in hash functions for the third, for which no efficient quantum algorithm is known.

Remark 1.2 Post-quantum cryptography versus quantum key distribution

The two responses to the quantum threat are often confused, and they differ in every respect that matters for deployment. Post-quantum cryptography runs on existing hardware and networks, can be deployed now, provides both key exchange and signatures, and rests on computational hardness assumptions. Quantum key distribution requires a quantum channel and, beyond the few hundred kilometres that direct fibre transmission allows as of 2025, either trusted intermediate nodes or satellite links, both of which exist, or, for end-to-end security without trusting the intermediate nodes, the repeaters of section 1.1; it provides key

> agreement only, not encryption, signatures, or authentication, so the data is still encrypted with a symmetric cipher and the classical channel is still authenticated by classical means; and its security rests on physics together with that authentication and with the correctness of the physical model of the devices. The standardization bodies responsible for public infrastructure have chosen post-quantum cryptography as the primary response. Quantum key distribution is a complementary option for specific links where the infrastructure is justified.

1.3.4 Current hardware

No cryptographically relevant quantum computer exists at the time of writing, and this course does not predict when one will. The following example compares what the algorithm requires with what has been built.

Example 1.1 (Resource estimates for RSA-2048)

Figures as of 2025; to be reviewed before each delivery.

Item	Physical qubits	Time or scale
Shor's algorithm on RSA-2048, Gidney and Ekerå (2021) [11], surface code, stated error rates	about 20 million noisy qubits	about 8 hours
Shor's algorithm on RSA-2048, Gidney (2025) [12], revised layout and stated assumptions	under one million noisy qubits	under one week
Largest publicly reported gate-based processors (for example IBM Condor, 1121 qubits, announced December 2023 [13])	on the order of 1000 physical qubits	no error correction across the device
Largest logical-qubit demonstrations (for example Bluvstein et al., Nature, 2024 [14])	hundreds of physical qubits	tens of logical qubits, shallow circuits

The gap between the second row and the last two is about three orders of magnitude in qubit count, together with a gap in error rates and circuit depth that the table does not show. The estimates themselves fell by a factor of about 20 in four years through better algorithms and codes, which is why they carry a date. The table shows the distance between an algorithm and its deployment; it is not a prediction of when that distance closes.

The two statements that the lecture rejects are now both visible. The claim that everything is about to be broken ignores the table above and the symmetric ciphers that carry the traffic. The claim that there is no threat ignores Shor's polynomial cost, the harvest-now-decrypt-later argument, and the direction in which the estimates have moved. The response that follows from the evidence is a classical migration to post-quantum cryptography for public-key components, a review of key lengths for symmetric components, and continued monitoring of both algorithms and hardware. **??** explains what the physical qubits in the table are for and why so many are needed per logical qubit.

1.4 Summary

A quantum network distributes entanglement between nodes and uses it as a resource; it does not move classical data faster, and it cannot use classical repeaters because no-cloning forbids measuring or amplifying a qubit in transit. Entanglement swapping extends entanglement from two short links to one long one by a Bell measurement at the middle node and a Pauli correction at an end node, and it is teleportation applied to half of a Bell pair. A quantum repeater combines swapping with quantum memory and entanglement purification. Quantum key distribution turns measurement disturbance into eavesdropper detection; it needs an authenticated classical channel and classical postprocessing, and a network that delivers Bell pairs delivers keys. Shor's algorithm breaks RSA, Diffie-Hellman, and elliptic-curve cryptography at sufficient scale; Grover's algorithm only halves the effective key length of symmetric ciphers and hash functions, so AES-128 remains secure for the foreseeable future and AES-256 is far out of reach. The practical response is post-quantum cryptography, which is classical and deployable now; quantum key distribution is a complementary option that needs quantum infrastructure. No cryptographically relevant quantum computer exists at the time of writing, and neither imminent universal decryption nor the absence of any threat is an accurate description.

Homework 4, part B, can be attempted after this lecture.

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